

1. ROW_NUMBER()

Assigns a unique sequential number to each row.

```
SELECT
    employee_name,
    salary,
    ROW_NUMBER() OVER (ORDER BY salary DESC) AS
row_num
FROM employees;
```

****Use Case:**** Pagination, duplicate removal,
top-N records.

2. RANK()

Assigns the same rank for ties and skips the next rank(s).

```
SELECT
    employee_name,
    salary,
    RANK() OVER (ORDER BY salary DESC) AS
rank_no
FROM employees;
```

****Use Case:** Competition rankings,
leaderboards.**

3. DENSE_RANK()

Assigns the same rank for ties without skipping ranks.

```
SELECT
    employee_name,
    salary,
    DENSE_RANK() OVER (ORDER BY salary DESC) AS
dense_rank_no
```

```
FROM employees;
```

****Use Case:**** Top customers, top-selling products.

4. LAG()

Returns a value from the previous row.

```
SELECT  
    transaction_date,  
    amount,
```

```
    LAG(amount) OVER (ORDER BY transaction_date)
AS previous_amount
FROM transactions;
```

****Use Case:**** Compare current transaction with previous transaction.

5. LEAD()

Returns a value from the next row.

```
SELECT
```

```
transaction_date,  
amount,  
LEAD(amount) OVER (ORDER BY  
transaction_date) AS next_amount  
FROM transactions;
```

****Use Case:**** Compare current transaction with
next transaction.
